Chapter 3

Precision and Accuracy of Chemical Analyses

***Fundamentals of Analytical Chemistry*, Skoog, et al., 10th Ed., 2022**

3-1, 3-5, 3-7, 3-8, 3-9, 3-10, 3-12

KHP lab calculations

Titrant (NaOH) standardization

$$0.2572\ g\ KHP\ x\ \frac{1\ \ mol\ KHP}{204.23\ g\ KHP}\ x\ \frac{1\ mol\ NaOH}{1\ mol\ KHP}\ x\frac{1}{0.02401\ L} = 0.05240\ M\ NaOH$$

Unknown Calculations

Use the mean of the standardization results for the molarity of the titrant

Percentage of KHP in a sample

$$26.15\ mL\ x\ \frac{0.05240\ mmol\ NaOH}{mL}\ x\ \frac{1\ mmol\ KHP}{1\ mmol\ NaOH}\ x\ \frac{204.23\ mg\ KHP}{1\ mmol\ KHP}\ x\ \frac{100\%}{498.7\ mg} = 56.12\ \%$$

Error: difference between accepted and measured value

Also used to describe uncertainty in the measurement

Replicates: measurements carried out in exactly the same way on the same sample.

N_i = number of measurements

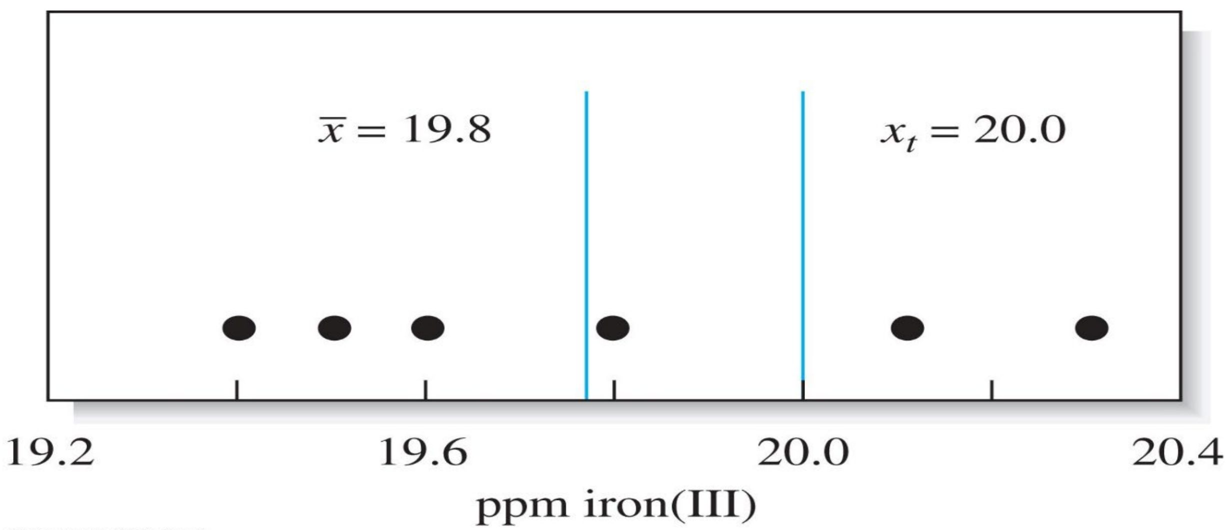
The most widely used measure of the central value is the **mean, x**

aka the arithmetic mean or the average is obtained by dividing the sum of replicate measurements by the number of measurements

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

The **median** is the middle result when replicate data are arranged in increasing (or decreasing) order. For an even number of results the average of the two middle results is used.

Example 3-1



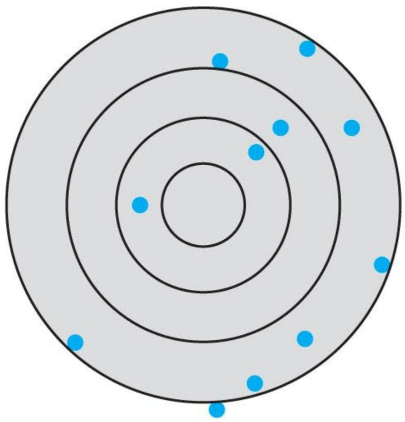
x_t = true or accepted value

$$\bar{x} = \frac{19.4 + 19.5 + 19.6 + 19.8 + 20.1 + 20.3}{6} = 19.78 = 19.8\ ppm\ Fe$$

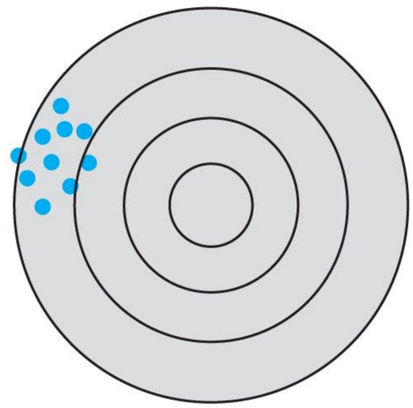
$$median = \frac{19.6+19.8}{2} = 19.7\ ppm\ Fe$$

Precision describes the reproducibility of measurements *(the closeness of results that have been obtained in exactly the same way)*

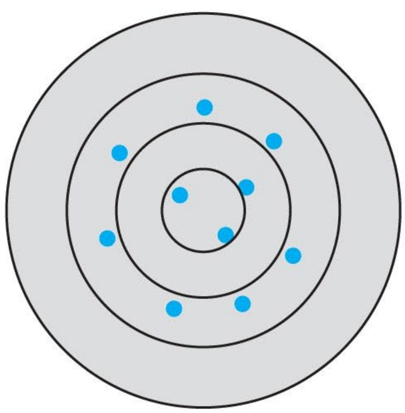
Accuracy indicates the closeness of the measurement to the true value and is expressed as error.



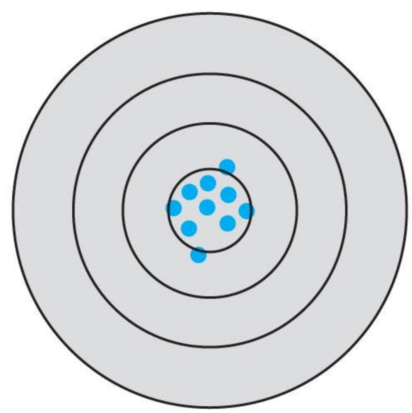
Low accuracy, low precision



Low accuracy, high precision



High accuracy, low precision



High accuracy, high precision

Absolute error E of a quantity x is given by

(*absolute here is different than in mathematics in that it can have a negative sign*)

$$E = x_i - x_t$$

Can be positive or negative

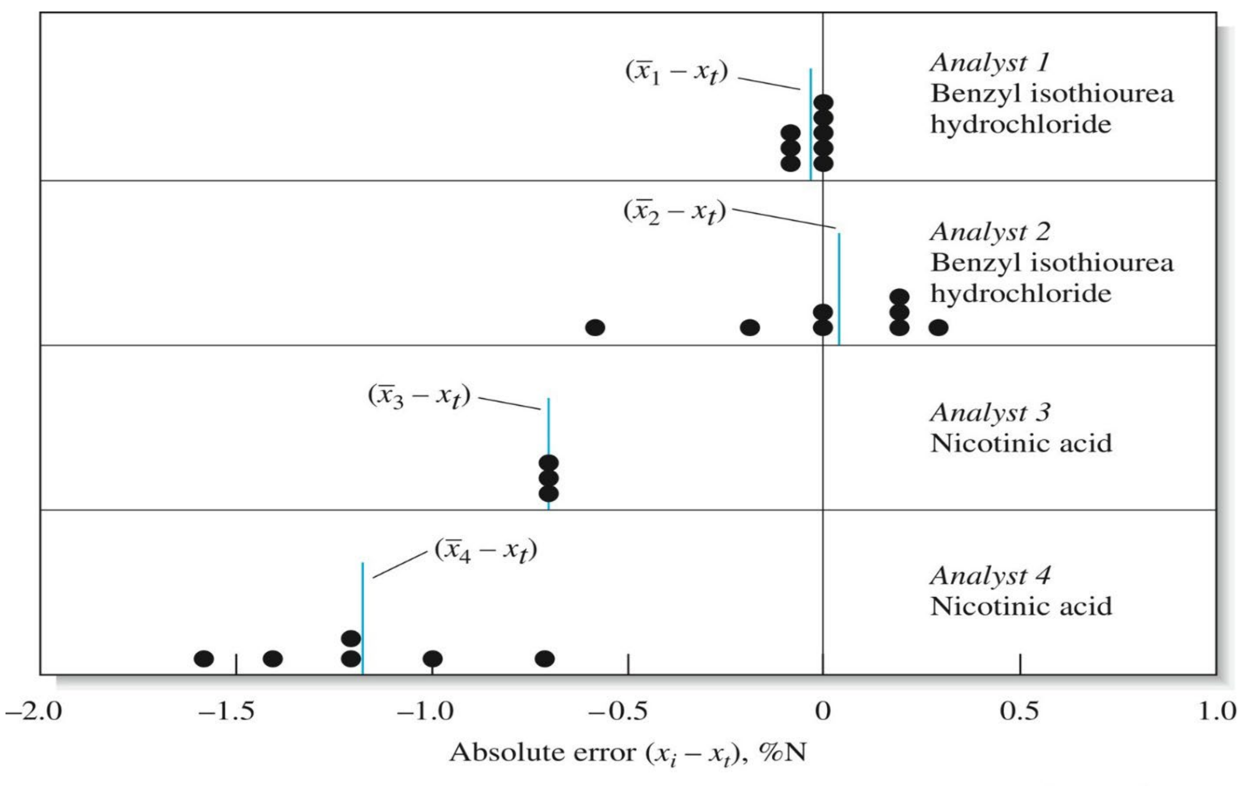
Relative Error E_r is often a more useful quantity than the absolute error.

$$E_r = \frac{x_i - x_t}{x_t} x\ 100\%$$

$$E_r = \frac{19.8 - 20.0}{20.0} x\ 100\% = -1\%$$ **(note the sign is negative)**

Types of Errors

Results can be precise without being accurate and accurate without being precise. The danger is in assuming that precise results are also accurate.



(Data from C. O. Willits and C. L. Ogg, *J. Assoc. Offic. Anal. Chem.*, **1949**, *32*, 561.)

Random (or indeterminate) error causes the data to be scattered symmetrically around a mean value (affects precision).

Gross errors are often large errors which result from human error. *Can be high or low eg. losing ppt or touching weigh bottles after its empty mass is determined.*

Outliers result from gross error.

Systematic (or determinate) error causes the mean to differ from the accepted value) (affects accuracy). Systematic errors have a definite value, an assignable cause, and are of the same magnitude for replicate measurements. They lead to a bias (negative if results are low or positive if values are high)

<u>**Sources of Systematic Errors**</u>

Instrumental Errors -malfunctioning instrument, faulty calibration of pipettes vol. flasks, leaky burets, varying lab temperature (throws off volumes dispensed*)*

Method Errors non-ideal chemical or physical behavior of analytical systems, reagent problems (*think of titrations and adding excess to get a color change, incomplete titration, interferences, side reactions*)

Personal Errors (carelessness, personal limitations ie color blindness, reading scales, bias towards 0 and 5 for scales)

Systematic Errors on Analytical Results

Can be either constant or proportional.

Constant errors stay the same as the size of the quantity measured increases. The effect of a constant error becomes more serious as the quantity measured decreases.

Absolute error is fixed.

Relative error can change with the sample size, can be minimized by increasing sample size.

Example 3-2

1. Suppose 0.05 mg of ppt is lost as a result of being washed with 200 ml of wash liquid. If the ppt weights 500 mg, the relative error due to solubility loss is

2. Loss of the same quantity (0.05 mg) out of a total 50 mg of ppt results in a relative error =

Proportional Errors increase as a fixed percentage of the sample

Usually from interferences (if you get a % over 100%)

Changing sample size does not help

Detection of Systematic Method Errors

Analysis of Standard Samples (standard reference materials)

These are often synthesized but may not reveal unexpected interferences

Available from NIST (national Institute of Standards and Technology)

Concentration of components is determined in one of three ways

1) Analysis with a previously validated reference method
2) Analysis by two or more independent, reliable measurement methods
3) Analysis by a network of co-operating laboratories that are technically competent and thoroughly knowledgeable.

Independent Analysis

Comparison of results from two independent methods. *(The methods should differ as much as possible)*.

Blank Determinations (*like a blank titration for the KHP lab*)

A blank contains the reagents and solvents but no analyte

Blank determinations reveal errors due to interferences and overestimations in titrations.

Variations in Sample Size

Can be used to detect constant errors.

READ THE CHAPTER!!!